MULTIMODAL FOOD FRESHNESS / STUDENT PROTOTYPE

FreshSense AI

A camera based food inspection prototype that reads package text, estimates visual freshness, and keeps the evidence together over repeated scans.

CORE WORKFLOW

Photo to evidence

Vision, OCR, product clues, storage context

STANDOUT IDEA

Freshness Passport

A timeline for the same tracked food item

PROTOTYPE STAGE

Phase 1

Working foundation; research models remain in progress

PROJECT REPORT

Prototype overview, AI workflow, novelty, limits, and live demo guide

A transparent prototype

FreshSense reports what its prototype actually observed. It distinguishes package text from user input, flags partial expiry stamps, and explains when evidence is missing. It is a demonstration and research platform, not a certified food safety test.

01 Project overview

FreshSense AI combines food photo analysis with package text and storage context. The aim is to make a scan understandable: the result should show the cues behind an assessment, what the system could not read, and how observations change when the same item is scanned again.

What happens during a scan

01 / CAPTURE	02 / READ	03 / ASSESS	04 / RECORD
Take or upload a food image.	Vision features and OCR inspect the image and printed label.	Expiry, vision and available sensor context feed deterministic rules.	A Freshness Passport stores evidence and the scan timeline.

AI and prototype capabilities

Capability	What the prototype does	Evidence shown
Food vision	Uses interpretable image features and a prototype model path.	Model type and prototype or model score; scores are not calibrated probabilities.
Expiry OCR	Reads package text and parses labels such as best before or expiry when visible.	OCR provider, recognized text, label cue, and full or partial date status.
Product category	Uses package words or a user supplied label to identify categories.	Category source, matched words, and match strength when available.
Evidence rules	Combines available signals with deterministic assessment rules.	Used, missing, warning, or conflicting evidence.
Repeat scan record	Groups scans under a named item for a chronological view.	Freshness Passport with scan history and storage context.

Why the label readout matters

A stamp like '04 JUL' may be visible while its year is not. FreshSense can retain that partial cue for inspection, but it avoids inventing a year or using an incomplete date as a full expiry calculation.

02 Novelty and research value

The project's clearest distinctive contribution is its **Longitudinal Freshness Passport**: scans of the same named food item become an evidence timeline instead of isolated scores. The record can preserve visual estimates, printed date cues, storage context, warnings, missing inputs, and conflicts between signals.

A testable research question

Does keeping repeated, same item observations and their evidence improve traceability and usefulness compared with isolated image only scans? With a suitable paired dataset, a second question is whether expiry and calibrated sensor evidence improve held out freshness classification over the image only baseline.

Current evidence and limits

Area	Current project status
Vision model	Prototype output is clearly labeled. Repository notes report an exploratory image candidate at 52.6% accuracy on 445 held out AgriFreshNET images; it was not promoted because performance was too weak.
Multimodal fusion	Deterministic rules are implemented as a foundation. There is no paired dataset result showing that multimodal fusion improves accuracy.
Expiry reading	Date parsing and OCR provider support are present. Reading quality depends on image clarity, stamp orientation, OCR availability, and whether the package includes a readable year.
Sensors and shelf life	Sensor interface is present; hardware calibration and trained shelf life prediction remain future work.

Next experiments

- Collect repeated photos of the same item over controlled storage times, with human reviewed freshness labels and capture conditions.
- Calibrate temperature, humidity, and VOC equipment before treating values as measurements; mark simulated values as demo estimates.
- Split data by food item, batch, and capture session to avoid near duplicate images crossing train and test sets.
- Compare image only and Passport assisted systems on held out item groups; report per class recall, macro F1, and uncertainty.
- Review related research before claiming academic novelty; the Passport is a project contribution and research hypothesis, not proven priority.

Use responsibly

Freshness outputs are informational. They do not certify food safety and should not override package instructions, proper storage, or a person's judgement.

03 Demo and phone guide

The prototype has a React/Vite web interface, a FastAPI backend, a local SQLite demo option, and a PostgreSQL configuration path. A seeded local demo account is documented for classroom walkthroughs.

Run on the PC

Step	Action
1	Install the project dependencies described in README.md (Node.js/npm and Python packages).
2	For the local demo, use backend/.env with the SQLite example configuration. Keep secrets local.
3	Run start-dev.cmd from the project folder to start the web app and API.
4	Open the local address printed in the launcher. Sign in with the demo credentials listed in README.md.
5	Open New scan, take or upload a package photo, inspect the result, and optionally name a Passport item for repeat scans.

Open it on a phone

- Connect the phone and PC to the same trusted Wi-Fi network.
- Double click **start-phone.cmd** on Windows and keep its window open. It prints the PC address to use.
- Open that printed address in the phone browser, sign in, then go to **New scan / Take photo**. The app uses the phone's rear camera capture flow and sends the image to the PC backend.
- If the page cannot load, allow Node/Python through Windows Firewall for private networks only, then retry. A guest Wi-Fi network may block device to device traffic.

Suggested teacher walkthrough

Show	Explain
Expiry label scan	The OCR reader exposes the exact recognized text and distinguishes complete dates from partial stamp fragments.
Package category	Matched words provide inspectable evidence for a category; a manually supplied label is identified as user input.
Freshness Passport	Repeated scans of one named item form a timeline with evidence provenance and conflicts.
Prototype limits	Show what is missing or estimated, and explain that reliable model claims require more paired data and evaluation.

Project references

Repository documentation consulted: README.md; ARCHITECTURE.md; NOVELTY.md; PROJECT_STATUS.md; PHASES.md; DESIGN.md. Status statements reflect those project notes and the implemented scan/result workflow; verify local launch and phone connectivity on the demo PC before presentation.

04 Literature and novelty check

A targeted literature search was conducted for expiry-date OCR, mobile food recognition, food inventory applications, and multimodal freshness sensing. This is a focused check, not a systematic review or a patent clearance search.

Assessment

The broad idea is not novel by itself. Published systems already cover expiry-date recognition, mobile OCR plus food image classification, household food inventory, and image plus sensor freshness assessment. FreshSense's strongest candidate distinction is its same-item timeline that preserves evidence source, missing signals, and disagreements across repeat scans. The search does not establish that this combination is unique.

Overlap with prior work

Research area	What the literature already demonstrates	Implication for FreshSense
Expiry-date OCR	Generalized package-date recognition and mobile date readers handle varied formats and difficult printing [1, 2].	Reading dates is established; robustness and truthful handling of partial reads matter.
Mobile scan workflow	A 2025 prototype combines mobile OCR and on-cloud product classification, with category-based shelf-life fallback [3].	The scan-and-classify combination is not new on its own.
Image plus sensors	Studies have combined freshness images with gas or electronic-nose, storage, and laboratory measurements [4, 5].	Multimodality is an established research direction; FreshSense needs calibrated inputs and a fair benchmark.
Household food tracking	Food inventory and expiry-management applications have been studied as household waste decision-support tools [6].	Tracking food and dates alone is not a novelty claim.

Defensible novelty statement

Suggested presentation wording

'FreshSense is a prototype that brings package OCR, image-based freshness cues, storage context, and a repeat-scan evidence timeline into one inspectable workflow. Its research question is whether preserving evidence provenance across scans improves traceability and usefulness; this remains to be evaluated.'

Can it be described as 100% efficient and error-free?

No. The project notes report an exploratory vision candidate at 52.6% accuracy on 445 held-out images and say it was not promoted. The current system is explicitly a prototype, and the repository has no paired dataset demonstrating multimodal gains. A credible report should say that FreshSense is a functioning prototype with an evidence-aware design, while accuracy and reliability remain to be measured on representative phone photos. Report OCR exact-match rate, date parsing accuracy, category precision/recall, end-to-end completion rate, latency, and error cases before making performance claims.

The goal can be high reliability; 100% should be presented only if a defined test set and independent evaluation actually support it. Even strong expiry OCR and freshness classification do not certify food safety.

05 Literature references

References selected for direct overlap with the project's features. DOI links are clickable.

[1] Seker, A. C., and S. C. Ahn. (2022). A generalized framework for recognition of expiration dates on product packages using fully convolutional networks. *Expert Systems with Applications*, 203, 117310. Supports that package expiry detection and date parsing are established tasks, including varied formats. https://doi.org/10.1016/j.eswa.2022.117310

[2] Peng, H., J. Bayon, J. Recas, and M. Guijarro. (2025). Efficient Expiration Date Recognition in Food Packages for Mobile Applications. *Algorithms*, 18(5), 286. Describes mobile package-date localization, OCR, and date parsing. https://doi.org/10.3390/a18050286

[3] Dospinescu, O., G.-L. Grigorcea, and B.-I. Lefter. (2025). A Mobile Approach to Food Expiration Date Determination Using OCR and On-Cloud Image Classification. *Applied System Innovation*, 8(6), 176. Directly overlaps mobile OCR plus food image classification. https://doi.org/10.3390/asi8060176

[4] Bhuiyan, Z. W., S. A. R. Haider, A. Haque, M. R. Uddin, and M. Hasan. (2024). IoT Based Meat Freshness Classification Using Deep Learning. *IEEE Access*, 12, 196047-196069. Combines camera-based classification and gas-sensor measurements. https://doi.org/10.1109/ACCESS.2024.3520029

[5] Senge, J. M., F. Kaltenecker, and C. Krupitzer. (2025). Integrating Sensor Data, Laboratory Analysis, and Computer Vision in Machine Learning-Driven E-Nose Systems for Predicting Tomato Shelf Life. *Chemosensors*, 13(7), 255. Combines VOC, image/color, weight, and storage-condition evidence; the study also discusses limits to generalization. https://doi.org/10.3390/chemosensors13070255

[6] Mastorakis, G., I. Kopanakis, J. Makridis, C. Chroni, K. Synani, K. Lasaridi, K. Abeliotis, I. Louloudakis, I. N. Daliakopoulos, and T. Manios. (2024). Managing Household Food Waste with the FoodSaveShare Mobile Application. *Sustainability*, 16(7), 2800. Describes a household food inventory and waste-management mobile app. https://doi.org/10.3390/su16072800

[7] Chen, Z., G. Zhang, and F. Zhang. (2026). Multimodal AI for Real-Time Food Safety and Quality: From Sensors to Foundation Models, Edge Deployment, and Regulation. *Food Science & Nutrition*, 14(2), e71534. Recent review of multimodal food-quality and freshness systems and their validation needs. https://doi.org/10.1002/fsn3.71534

Novelty conclusion

FreshSense has a clear project contribution and a defensible research direction, but this targeted search does not support a claim of first-of-its-kind technology. A stronger future contribution would be a publicly documented, grouped, repeat-scan dataset and an evaluation showing whether the evidence timeline improves traceability or decisions over isolated scans.